SAVEETHA INSTITUTE OF MEDICAL AND TECHNICAL SCIENCES

SIMATS ENGINEERING

SUB CODE & NAME: CSA17- ARTIFICIAL INTELLIGENCE

Exp 1:Write the python program to solve 8-Puzzle problem

CODE:

```
from collections import deque

start = (1, 2, 3,
         4, 0, 6,
         7, 5, 8)

goal = (1, 2, 3,
        4, 5, 6,
        7, 8, 0)

def solve(start, goal):
    queue = deque([(start, [])])
    visited = {start}

    while queue:
        state, path = queue.popleft()

        if state == goal:
            return path + [state]

        zero = state.index(0)
        r, c = divmod(zero, 3)
```

```
        for dr, dc in [(-1, 0), (1, 0),
                       (0, -1), (0, 1)]:
            nr, nc = r + dr, c + dc

            if 0 <= nr < 3 and 0 <= nc < 3:
                nz = nr * 3 + nc
                new_state = list(state)
                new_state[zero], new_state[nz] = (
                    new_state[nz], new_state[zero]
                )
                new_state = tuple(new_state)

                if new_state not in visited:
                    visited.add(new_state)
                    queue.append(
                        (new_state, path + [state])
                    )

    return None

solution = solve(start, goal)

if solution:
    for step, state in enumerate(solution):
        print("Step", step)
        for i in range(0, 9, 3):
            print(state[i:i+3])
        print()
```

else:

    print("No solution found")

OUTPUT:

